

Boundary-regular isometric symbols on analytic Besov spaces

Candidate partial result for arXiv:2207.12634

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Status. Candidate partial result, likely valid. The unrestricted conjecture for an arbitrary analytic symbol remains open in this packet. We prove it when the symbol extends holomorphically past the closed unit disk.

1 Source conjecture

For $1 < p < \infty$, the analytic Besov space B_p consists of the analytic functions on $\mathbb{D}$ for which

$$\|f\|_p^p = \int_{\mathbb{D}} |f'(z)|^p (1 - |z|^2)^{p-2} dA(z) < \infty,$$

with Banach norm $|f(0)| + \|f\|_p$. Allen, Heller, and Pons [1] proved that rotations are the only isometric composition symbols for $1 < p < 2$, and obtained several reductions for $p > 2$. They ask whether the same conclusion holds without an extra hypothesis when $p > 2$.

univalent function, then φ is a rotation of the disk.

For a general isometry we are only able to obtain a partial result which relaxes the hypothesis of the analogous statement given in [14].

Theorem 3.9. *Let $p > 2$ and suppose that $n_\varphi(w) = 1$ a.e. in some neighborhood of the origin. Then C_φ is an isometry on B_p if and only if φ is a rotation of the disk.*

Proof. If φ is a rotation of the disk, then it is clear that C_φ is an isometry on B_p . Assuming that $n_\varphi(w) = 1$ a.e. in some neighborhood of the origin it follows that there is a neighborhood U of $\varphi(0) = 0$ such that $n_\varphi(w) = 1$ a.e. on U . Since φ is continuous, $\varphi^{-1}(U)$ is an open set and thus there is an $0 < r < 1$ such that $\varphi(D(0, r)) \subseteq U$, where $D(0, r)$ is the Euclidean disk centered at the origin with radius r . Set $E = \varphi(D(0, r))$; then φ^{-1} is well-defined on E as a single-valued inverse mapping and $\varphi^{-1}(E) = D(0, r)$. By Lemma 3.4 and Eqn. (2.2) we have (by techniques similar to the previous theorem)

$$\begin{aligned} \int_E (1 - |z|^2)^{p-2} dA(z) &= \int_{D(0, r)} |\varphi'(z)|^p (1 - |z|^2)^{p-2} dA(z) \\ &\leq \int_{D(0, r)} |\varphi'(z)|^2 (1 - |\varphi(z)|^2)^{p-2} dA(z) \\ &= \int_E (1 - |w|^2)^{p-2} dA(w). \end{aligned}$$

This computation implies that equality must hold throughout and as in the proof of Theorem 3.6, it must be the case that φ is a rotation of the disk. $\square$

We conjecture that every isometric composition operator on B_p with $p > 2$ must be induced by a rotation of the disk but the techniques we have employed seem unable to support this claim. Perhaps more sophisticated methods such as those used in [9] will prove more fruitful.

The theorem below answers the question for symbols with analytic boundary regularity. In particular it covers rational self-maps with no pole on $\overline{\mathbb{D}}$.

2 Idea of the proof

The isometry condition supplies two rigid pieces of information: the symbol is full up to area zero, and the hyperbolic derivative weights over every fiber sum to one. Analytic continuation past the circle turns fullness into the boundary identity $|\varphi| = 1$, so the symbol is a finite Blaschke product. Near a boundary value, every inverse branch of such a product has hyperbolic derivative tending to one. The fiber identity then says that the degree of the product is one. The remaining normalization $\varphi(0) = 0$ makes the resulting automorphism a rotation.

3 The fiber identity

Fix $p > 2$ and write $\beta = p - 2 > 0$. If C_φ is an isometry on B_p , the source proves that

$$\varphi(0) = 0, \quad A(\mathbb{D} \setminus \varphi(\mathbb{D})) = 0, \quad (1)$$

and that $W_{\varphi', \varphi}$ is an isometry on the weighted Bergman space A_{β}^p ; see [1, Lemmas 3.1 and 3.4]. Kolaski's Borel equimeasurability identity, followed by the area formula, therefore gives

$$\sum_{\varphi(z)=w} \left(\frac{|\varphi'(z)|(1-|z|^2)}{1-|w|^2} \right)^{\beta} = 1 \quad (2)$$

for almost every regular value $w \in \mathbb{D}$. Each summand is at most one by the Schwarz–Pick lemma. Formula (2) is the only isometric input needed after (1).

4 Full boundary-regular maps

Lemma 1. *Let $\varphi : \mathbb{D} \rightarrow \mathbb{D}$ extend holomorphically to a neighborhood of $\overline{\mathbb{D}}$. If $A(\mathbb{D} \setminus \varphi(\mathbb{D})) = 0$, then $|\varphi(\zeta)| = 1$ for every $\zeta \in \mathbb{T}$. Consequently φ is a finite Blaschke product up to a unimodular constant.*

Proof. Polar Fubini applied to the null set $\mathbb{D} \setminus \varphi(\mathbb{D})$ shows that for almost every $\eta \in \mathbb{T}$ there are radii $r_n \uparrow 1$ with $r_n \eta \in \varphi(\mathbb{D})$. Choose $z_n \in \mathbb{D}$ with $\varphi(z_n) = r_n \eta$. A subsequence converges in $\overline{\mathbb{D}}$ to some ζ . Continuity gives $\varphi(\zeta) = \eta$. The point ζ cannot lie in $\mathbb{D}$, by the maximum-modulus principle, so $\zeta \in \mathbb{T}$.

Thus the image under $\varphi|_{\mathbb{T}}$ of

$$Z = \{\zeta \in \mathbb{T} : |\varphi(\zeta)| = 1\}$$

contains almost all of $\mathbb{T}$. The boundary map is Lipschitz. Hence Z cannot have arc-length measure zero, since Lipschitz maps send null sets to null sets. The function $t \mapsto |\varphi(e^{it})|^2 - 1$ is real analytic and vanishes on the positive-measure set corresponding to Z ; it therefore vanishes identically. The standard zero-factorization and maximum principle now show that φ is a finite Blaschke product times a unimodular constant. $\square$

5 Rotation theorem

Theorem 2. *Let $p > 2$, let $\varphi : \mathbb{D} \rightarrow \mathbb{D}$ extend holomorphically to a neighborhood of $\overline{\mathbb{D}}$, and suppose C_φ is an isometry on B_p . Then there is $\theta \in \mathbb{R}$ such that*

$$\varphi(z) = e^{i\theta} z \quad (z \in \mathbb{D}).$$

Proof. By (1) and Lemma 1, φ is a finite Blaschke product; let its degree be $d \geq 1$. Choose a point $\eta \in \mathbb{T}$ and a sequence $w_n \rightarrow \eta$ such that every w_n is a regular value and (2) holds. This is possible because the set on which (2) holds has full area and is therefore dense, and a finite Blaschke product has only finitely many critical values.

For large n , the equation $\varphi(z) = w_n$ has d simple roots $z_{1,n}, \dots, z_{d,n}$. After labeling and passing to subsequences, each $z_{j,n}$ tends to a boundary point ζ_j with $\varphi(\zeta_j) = \eta$. A finite Blaschke product has a finite nonzero angular derivative at every point of $\mathbb{T}$. The Julia–Carathéodory identity, applied along the inverse branch, gives

$$\frac{|\varphi'(z_{j,n})|(1 - |z_{j,n}|^2)}{1 - |w_n|^2} \rightarrow 1 \quad (j = 1, \dots, d).$$

Passing to the limit in (2) yields

$$1 = \lim_{n \rightarrow \infty} \sum_{j=1}^d \left(\frac{|\varphi'(z_{j,n})|(1 - |z_{j,n}|^2)}{1 - |w_n|^2} \right)^\beta = d.$$

Thus $d = 1$, so φ is a disk automorphism. Finally $\varphi(0) = 0$ by (1); hence it is a rotation. □

6 What remains

The boundary extension is used twice: to turn fullness into the inner boundary condition, and to prevent fiber mass from escaping through infinitely many irregular sheets. The proof does not extend to an arbitrary analytic symbol by approximation. Indeed, at a point where $|\varphi'| = 1$ for a hypothetical nonrotation fixing zero, its local term in (2) is strictly less than one, so other sheets are compulsory. Those sheets may run to the boundary with nonvanishing hyperbolic derivative.

A later dissertation and article claim the unrestricted theorem [2, 3]. The available dissertation proof suppresses the other sheets after choosing a local univalence disk. A companion proof-gap packet records that issue; it is not a counterexample to the theorem. Thus Theorem 2 should be read as a rigorous partial result, not as a replacement claim for the unrestricted classification.

References

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